LINUS LIU

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EDUCATION

Rice University Houston, TX | Expected May 2028
B.S. Computer Science; B.A. Computational and Applied Mathematics
Mathematics Major GPA: 4.00/4.00
Coursework: Machine Learning; Deep Learning; Algorithms; Honors Linear Algebra and Multivariable Calculus; Statistics for Generative Models.

RESEARCH INTERESTS & SKILLS

Interests: Mixture-of-experts (MoE), AI for science, and efficient large-scale AI
Tools: Python, PyTorch, JAX, NumPy, pandas, scikit-learn, SQL, HDF5
Methods: Controlled experiments, ablation studies, benchmark reproduction, retrieval and latency evaluation

SELECTED RESEARCH EXPERIENCE

MD ANDERSON CANCER CENTER Mar – Sept 2026
Researcher | Computational Pathology | Advisor: Andrew Song

- Co-developed a **whole-slide pathology pipeline**: tissue segmentation, patch extraction, **CONCH/Virchow2** encoding, and **CLAM-compatible HDF5** feature bags for downstream modeling.
- Reproduced **HEST** gene-expression prediction benchmarks using **256-D PCA and Ridge regression**, matching published Pearson-correlation baselines across **three cancer cohorts**.
- Developing a **JEPA-style latent world model** to predict future pathology embeddings conditioned on actions and elapsed time; exploring heatmaps and nearest-slide retrieval to visualize progression.

GOOGLE DEEPMIND Feb – Mar 2026
Research Extern | LLM Self-Evolving Memory | Remote

- Developed hypotheses on **self-evolving LLM memory**, reviewing continual-learning research to investigate information retention and retrieval across sessions.
- Designed **controlled experiments** on memory retention and retrieval accuracy across context lengths, collaborating with research leads to refine and test hypotheses.

LENOVO Apr – Aug 2025
Machine Learning Intern | Sequential Decision Modeling | Beijing, China

- Implemented a **Decision Pre-trained Transformer** for multi-zone HVAC control on IoT data; built reproducible **Python/JAX training and offline evaluation workflows**.
- Ran **context-length and regularization ablations**; used SQL filtering, metric analysis, and replay-based testing to investigate failure cases and model stability.

RICE UNIVERSITY Nov 2025 – Feb 2026
Research Assistant, Machine Learning | Biosignal Gesture Recognition | Advisor: Prof. Momona Yamagami

- Built a **PyTorch** gesture-recognition pipeline for **88-channel EMG and IMU data**, combining a **1D CNN** with a **two-layer LSTM** for upper-body movement classification.
- Implemented a **PCA/Euclidean-distance template-matching baseline**; ran grid searches over kernel sizes, hidden layers, and dropout to investigate model convergence.

SELECTED APPLIED MACHINE LEARNING EXPERIENCE

WISHEE Dec 2025 – Present
Intern | Voice Assistant Retrieval | Shenzhen / Remote

- Built a retrieval service over **30,000 documents and 600K chunks**, serving **20 queries/sec** at **600 ms p95 first-token latency** for a smart-earbud voice assistant.
- Cached speculative searches on partial speech transcripts, cutting post-speech wait from **680 ms to 440 ms**.
- Combined dense vectors, **BM25**, metadata filtering, and reciprocal rank fusion, improving **recall@5 from 0.68 to 0.89** on **200 hand-labeled voice queries**.

SELECTED HONORS

S.-T. Yau High School Science Award – Third Prize (2023); **HiMCM** – Meritorious Award (2022).
Chinese Chemistry Olympiad – Provincial Third Prize (2022).